

Python decorators for adding function to function

https://www.w3schools.com/PYTHON/python_decorators.asp

Python typing - types

<https://docs.python.org/3/library/typing.html>

Due Friday 5/8

1. Build an outline somewhere
2. Code
3. Paragraph on complexities like big o

PYTHON

Basically just the movie project but plan it out better and make it keep track of things over time

THE PLAN

Use one csv file for output of whole thing on one

Stock.csv:

Item name | Current Number In Stock

Use list of tuples {string : int[]}] to store {item name : stock_record[oldest stock record to most recent stock record]}

Binary search for item

Merge sort items

Predict run out using burn out rates

Juicemind movie store project as example

Data container: List of tuples (string, int[oldest -> newest stock value])

Class (python abstract base class) InventoryOperations - an interface

Use python decorators to make this abstract and a property. Property makes this a variable instead of a method

data()

defines what contains all of the data

load_inventory(csv file)

catch if the file does not exist

if it does exist then for each row in the file record the past items and their inventory

history

search(item: str) - > integer index or nothing

binary search.

start value and end value

find the middle of the start and end, if the middle value's name is equal to the item you are looking for then return that index

if the value is less than, then go to the right side

if alpha value is greater than, then go to the left side

if nothing is found return none

add(name : str, value : int):

use premade binary search algorithm to find if this exists

if it does exist then add to that value

if it doesn't exist, use another binary search to find insertion point but change the bounds throughout the function to allow for it to locate the correct position

calculatePrediction(name : str):

#use binary search function to find that item

check if there is enough data

#(item[0] - item[-1])/len(item) - calculate average roc

average roc is burn rate, calculate predicted periods remaining by the current value over burn rate

remove_item(name : str):

use pre-made binary search method to find index of item

remove tuple at that index

print that it was removed "{name} removed\n"

see_current_Stock(name: str):

#binary search and print item price history

```
update_Stock(name: str, change: int):  
    # use search to find item index  
    # update integer value
```

```
get_Inventory():  
    #for item in list:  
        write to csv "{item[0]}, {item[1]}"
```

```
main():  
    try:  
        #get the list and all that, give user 8 choices:  
        add, update, view, remove, predict, display all, save, and exit  
    Except Exception e:  
        print("Error occurred")
```

Old Movie project as reference

<https://play.juicemind.com/sandbox/ACeqkgeEokj1YWnha2vp>

Codehs;

<https://codehs.com/sandbox/sunl1ght/crud-app-project?collaborate=-Os7g4ewTeNRljyqYaMM>